

Q1 JEE Main 2020 - 7 Jan (Morning)

If $\operatorname{Re}\left(\frac{z-1}{2z+i}\right) = 1$, where $z = x + iy$, then the point (x, y) lies on a:

- (A) Straight line with slope 2
- (B) Straight line with slope $-\frac{1}{2}$
- (C) circle with diameter $\frac{\sqrt{5}}{2}$
- (D) circle with diameter $\frac{1}{2}$

Q2 JEE Main 2020 - 7 Jan (Evening)

If $\frac{3+i\sin\theta}{4-i\cos\theta}$, $\theta \in [0, 2\pi]$, is a real number, then an argument of $\sin\theta + i\cos\theta$ is:

- (A) $-\tan^{-1}\frac{3}{4}$
- (B) $\pi - \tan^{-1}\frac{3}{4}$
- (C) $\pi - \tan^{-1}\frac{4}{3}$
- (D) $\tan^{-1}\frac{4}{3}$

Q3 JEE Main 2020 - 8 Jan (Morning)

If the equation, $x^2 + bx + 45 = 0$ ($b \in R$) has conjugate complex roots and they satisfy $|z + 1| = 2\sqrt{10}$, then :

- (A) $b^2 + b = 12$
- (B) $b^2 - b = 30$
- (C) $b^2 - b = 36$
- (D) $b^2 + b = 30$

Q4 JEE Main 2020 - 8 Jan (Evening)

Let $a = \frac{-1+i\sqrt{3}}{2}$ is $a = (1 + \alpha) \sum_{k=0}^{100} \alpha^{2k}$ and $b = \sum_{k=0}^{100} \alpha^{3k}$, then a and b are the roots of the quadratic equation :

- (A) $x^2 - 102x + 101 = 0$
- (B) $x^2 - 101x + 100 = 0$
- (C) $x^2 + 101x + 100 = 0$
- (D) $x^2 + 102x + 100 = 0$

Q5 JEE Main 2020 - 9 Jan (Morning)

If $\left| \frac{z-i}{z+2i} \right| = 1$, $|z| = \frac{5}{2}$ then value of $|z + 3i|$ is

- (A) $\frac{7}{2}$
- (B) $\sqrt{10}$
- (C) $\sqrt{5}$
- (D) $\sqrt{3}$

Q6 JEE Main 2020 - 9 Jan (Evening)

If z be a complex number satisfying $|\operatorname{Re}(z)| + |\operatorname{Im}(z)| = 4$, then $|z|$ cannot be:

- (A) $\sqrt{10}$
- (B) $\sqrt{8}$
- (C) $\sqrt{7}$
- (D) $\frac{\sqrt{17}}{2}$

Q7 JEE Main 2020 - 2 Sep (Morning)

The value of $\left(\frac{1 + \sin \frac{2\pi}{9} + i \cos \frac{2\pi}{9}}{1 + \sin \frac{2\pi}{9} - i \cos \frac{2\pi}{9}} \right)^3$ is :

- (A) $-\frac{1}{2}(1 - i\sqrt{3})$
- (B) $\frac{1}{2}(1 - i\sqrt{3})$
- (C) $\frac{1}{2}(\sqrt{3} - i)$
- (D) $-\frac{1}{2}(\sqrt{3} - i)$

Q8 JEE Main 2020 - 2 Sep (Evening)

The imaginary part of $(3 + 2\sqrt{-54})^{1/2} - (3 - 2\sqrt{-54})^{1/2}$ can be

- (A) $\sqrt{6}$
- (B) $-\sqrt{6}$
- (C) $-2\sqrt{6}$
- (D) 6

Q9 JEE Main 2020 - 3 Sep (Morning)

If $\left(\frac{1+i}{1-i} \right)^{m/2} = \left(\frac{1+i}{i-1} \right)^{n/3} = 1$, $(m, n \in \mathbb{N})$ then the greatest common divisor of the least values of m and n is

Q10 JEE Main 2020 - 3 Sep (Evening)

If z_1, z_2 are complex numbers such that $\operatorname{Re}(z_1) = |z_1 - 1|$, $\operatorname{Re}(z_2) = |z_2 - 1|$, and $\arg(z_1 - z_2) = \frac{\pi}{6}$, then $\operatorname{Im}(z_1 + z_2)$ is equal to

- (A) $\frac{\sqrt{3}}{2}$
- (B) $\frac{1}{\sqrt{3}}$
- (C) $\frac{2}{\sqrt{3}}$
- (D) $2\sqrt{3}$

Q11 JEE Main 2020 - 4 Sep (Morning)

Let $u = \frac{2z+i}{z-ki}$, $z = x + iy$ and $k > 0$. If the curve represented by $\operatorname{Re}(u) + \operatorname{Im}(u) = 1$ intersects the y-axis at the point P and Q where $PQ = 5$, then the value of K is :

- (A) $1/2$
- (B) $3/2$
- (C) 2
- (D) 4

Q12 JEE Main 2020 - 4 Sep (Evening)

If a and b are real numbers such that $(2 + \alpha)^4 = a + b\alpha$, where $\alpha = \frac{-1+i\sqrt{3}}{2}$, then $a + b$ is equal to

- (A) 33
- (B) 9
- (C) 24
- (D) 57

Q13 JEE Main 2020 - 5 Sep (Morning)

If the four complex numbers $z, \bar{z}, \bar{z} - 2\operatorname{Re}(\bar{z})$ and $z - 2\operatorname{Re}(z)$ represent the vertices of a square of side 4 units in the Argand plane, then $|z|$ is equal to :

- (A) $4\sqrt{2}$
- (B) 2
- (C) $2\sqrt{2}$
- (D) 4

Q14 JEE Main 2020 - 5 Sep (Evening)

The value of $\left(\frac{-1+i\sqrt{3}}{1-i}\right)^{30}$ is :

- (A) -2^{15}
- (B) -2^{15}
- (C) 2^{15}
- (D) 6^5

Q15 JEE Main 2020 - 6 Sep (Morning)

The region represented by $\{z = x + iy \in C : |z| - \operatorname{Re}(z) \leq 1\}$ is also given by the inequality:

- (A) $y^2 \leq x + \frac{1}{2}$
- (B) $y^2 \leq 2\left(x + \frac{1}{2}\right)$
- (C) $y^2 \geq x + 1$
- (D) $y^2 \geq 2(x + 1)$

Q16 JEE Main 2020 - 6 Sep (Evening)

Let $z = x + iy$ be a non-zero complex number such that $z^2 = i|z|^2$, where $i = \sqrt{-1}$, then z lies on the:

- (A) line, $y = x$
- (B) imaginary axis
- (C) real axis
- (D) line, $y = -x$

Answer Key

Q1 (C)

Q2 (C)

Q3 (B)

Q4 (A)

Q5 (A)

Q6 (C)

Q7 (D)

Q8 (C)

Q9 (4)

Q10 (D)

Q11 (C)

Q12 (B)

Q13 (C)

Q14 (A)

Q15 (B)

Q16 (A)

Q1

$$z = x + iy$$

$$\left(\frac{z-1}{2z+i} \right) = \frac{(x-1) + iy}{2(x+iy) + i}$$

$$= \frac{(x-1) + iy}{2x + (2y+1)i} \times \frac{2x - (2y+1)i}{2x - (2y+1)i}$$

$$\operatorname{Re} \left(\frac{z-1}{2z+i} \right) = \frac{2x(x-1) + y(2y+1)}{(2x)^2 + (2y+1)^2} = 1$$

$$\Rightarrow 2x^2 + 2y^2 - 2x + y = 4x^2 + 4y^2 + 4y + 1$$

$$\Rightarrow 2x^2 + 2y^2 + 2x + 3y + 1 = 0$$

$$\Rightarrow x^2 + y^2 + x + \frac{3}{2}y + \frac{1}{2} = 0$$

$$\text{Circle with centre } \left(-\frac{1}{2}, -\frac{3}{4} \right)$$

$$r = \sqrt{\frac{1}{4} + \frac{9}{16} - \frac{1}{2}} = \sqrt{\frac{4+9-8}{16}} = \frac{\sqrt{5}}{4}$$

Q2

$$z = \frac{(3 + i \sin \theta)}{(4 - i \cos \theta)} \times \frac{(4 + i \cos \theta)}{(4 + i \cos \theta)}$$

as z is purely real

$$\Rightarrow 3 \cos + 4 \sin \theta = 0$$

$$\Rightarrow \tan \theta = -\frac{3}{4}$$

$$\arg(\sin \theta + i \cos \theta) = \pi + \tan^{-1} \left(\frac{\cos \theta}{\sin \theta} \right)$$

$$= \pi + \tan^{-1} \left(-\frac{4}{3} \right) = \pi - \tan^{-1} \left(\frac{4}{3} \right)$$

Q3

Let $z = \alpha \pm i\beta$ be roots of the equation

$$\text{So } 2\alpha = -b \text{ and } \alpha^2 + \beta^2 = 45, (\alpha + 1)^2 + \beta^2 = 40$$

$$\text{So } (\alpha + 1)^2 - \alpha^2 = -5$$

$$\Rightarrow 2\alpha + 1 = -5 \Rightarrow 2\alpha = -6$$

$$\text{so } b = 6$$

$$\text{hence } b^2 - b = 30$$

Q4

$$\alpha = \omega, b = 1 + \omega^3 + \omega^6 + \dots = 101$$

$$a = (1 + \omega)(1 + \omega^2 + \omega^4 + \dots + \omega^{198} + \omega^{200})$$

$$= (1 + \omega) \frac{(1 - (\omega^2)^{101})}{1 - \omega^2} = \frac{(1 + \omega)(1 - \omega)}{1 - \omega^2} = 1$$

$$\text{Equation : } x^2 - (101 + 1)x + (101) \times 1 = 0$$

$$\Rightarrow x^2 - 102x + 101 = 0$$

Q5

$$x^2 + (y - 1)^2 = x^2 + (y + 2)^2$$

$$-2y + 1 = 4y + 4$$

$$6y = -3 \Rightarrow y = -\frac{1}{2}$$

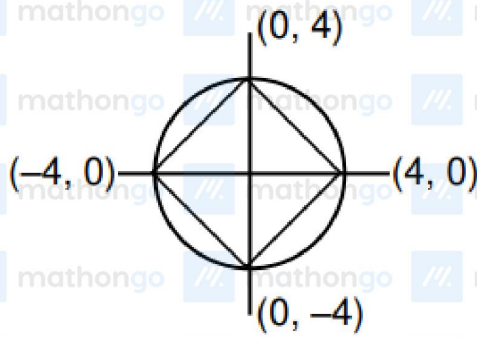
$$x^2 + y^2 = \frac{25}{4} \Rightarrow x^2 = \frac{24}{4} = 6$$

$$\Rightarrow z = \pm\sqrt{6} - \frac{i}{2}$$

$$|z + 3i| = \sqrt{6 + \frac{25}{4}} = \sqrt{\frac{49}{4}}$$

$$|z + 3i| = \frac{7}{2}$$

Q6



$$z = x + iy$$

$$|x| + |y| = 4$$

$$\text{Minimum value of } |z| = 2\sqrt{2}$$

$$\text{Maximum value of } |x| = 4$$

$$|z| \in [\sqrt{8}, \sqrt{16}]$$

So $|z|$ can't be $\sqrt{7}$

Q7

$$\text{Let } \theta = \frac{2\pi}{9}$$

$$\therefore \left(\frac{1 + \sin \frac{2\pi}{9} + i \cos \frac{2\pi}{9}}{1 + \sin \frac{2\pi}{9} - i \cos \frac{2\pi}{9}} \right)^3 = \left(\frac{1 + \cos \left(\frac{\pi}{2} - \theta \right) + i \sin \left(\frac{\pi}{2} - \theta \right)}{1 + \cos \left(\frac{\pi}{2} - \theta \right) - i \sin \left(\frac{\pi}{2} - \theta \right)} \right)^3$$

$$= \left(\frac{\cos \left(\frac{\pi}{4} - \frac{\theta}{2} \right) + i \sin \left(\frac{\pi}{4} - \frac{\theta}{2} \right)}{\cos \left(\frac{\pi}{4} - \frac{\theta}{2} \right) - i \sin \left(\frac{\pi}{4} - \frac{\theta}{2} \right)} \right)^3$$

$$= \left(e^{i \left(\frac{\pi}{4} - \frac{\theta}{2} \right)} \cdot e^{i \left(\frac{\pi}{4} - \frac{\theta}{2} \right)} \right)^3$$

$$= \left(e^{i \left(\frac{\pi}{2} - \theta \right)} \right)^3 = e^{i \left(\frac{3\pi}{2} - 3\theta \right)}$$

$$= -\sin 3\theta - i \cos 3\theta$$

$$= -\sin \frac{2\pi}{3} - i \cos \frac{2\pi}{3}$$

$$= -\frac{\sqrt{3}}{2} + \frac{i}{2} = -\frac{1}{2}(\sqrt{3} - i)$$

Q8

$$\text{Let } \sqrt{3 + 6\sqrt{6}i} = a + ib$$

$$\Rightarrow a^2 - b^2 = 3 \text{ and } ab = 3\sqrt{6}$$

$$\Rightarrow a^2 + b^2 = 15$$

$$\text{So, } a = \pm 3 \text{ and } b = \pm \sqrt{6}$$

$$\sqrt{3 + 6\sqrt{6}i} = \pm(3 + \sqrt{6}i)$$

Similarly, $\sqrt{3 - 6\sqrt{6}i} = \pm(3 - \sqrt{6}i)$

$$\operatorname{Im}(\sqrt{3 + 6\sqrt{6}i} - \sqrt{3 - 6\sqrt{6}i}) = \pm 2\sqrt{6}$$

Q9

$$\left(\frac{1+i}{1-i}\right)^{m/2} = \left(\frac{1+i}{i-1}\right)^{n/3} = 1$$

$$\left(\frac{(1+i)^2}{2}\right)^{m/2} = \left(\frac{(1+i)^2}{-2}\right)^{n/3} = 1$$

$$\Rightarrow i^{m/2} = (-i)^{n/3} = 1$$

$$m_{1\text{east}} = 8, n_{1\text{east}} = 12$$

$$\operatorname{GCD}(8, 12) = 4$$

Q10

$$\text{Let } z_1 = x_1 + iy_1, z_2 = x_2 + iy_2$$

$$\Rightarrow x_1 = \sqrt{(x_1 - 1)^2 + y_1^2}$$

$$\Rightarrow x_1^2 = x_1^2 + 1 - 2x_1 + y_1^2$$

$$\Rightarrow y_1^2 - 2x_1 + 1 = 0 \text{ (i)}$$

Similarly

$$y_2^2 - 2x_2 + 1 = 0 \text{ (ii)}$$

Now

$$\text{(i) - (ii), } (y_1 - y_2)(y_1 + y_2) = 2(x_1 - x_2)$$

$$\Rightarrow y_1 - y_2 = \frac{2(x_1 - x_2)}{y_1 + y_2} \Rightarrow \frac{y_1 - y_2}{x_1 - x_2} = \frac{2}{y_1 + y_2}$$

Now

$$\arg(z_1 - z_2) = \frac{\pi}{6}$$

$$\tan^{-1}\left(\frac{y_1 - y_2}{x_1 - x_2}\right) = \frac{\pi}{6}$$

$$\frac{y_1 - y_2}{x_1 - x_2} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \frac{2}{y_1 + y_2} = \frac{1}{\sqrt{3}}$$

$$y_1 + y_2 = 2\sqrt{3}$$

$$\text{So } r_M = (z_1 + z_2) = y_1 + y_2 = 2\sqrt{3}$$

Q11

$$u = \frac{2(x+iy)+i}{(x+iy)-ki} = \frac{2x+i(2y+1)}{x+i(y-k)}$$

$$\therefore \operatorname{Re}(u) = \frac{2x^2+(y-K)(2y+1)}{x^2+(y-K)^2}$$

$$\text{and } \operatorname{Im}(u) = \frac{-2x(y-K)+x(2y+1)}{x^2+(y-K)^2}$$

$$\text{Also } \operatorname{Re}(u) + \operatorname{Im}(u) = 1$$

$$\Rightarrow 2x^2 + 2y^2 - 2Ky + y - k - 2xy + 2Kx + 2xy + x = x^2 + y^2 + K^2 - 2Ky$$

$$\text{Let } y_1 \& y_2 \text{ are roots of equations if } x = 0 \Rightarrow y^2 + y - K(K+1) = 0 \begin{cases} y_1 \\ y_2 \end{cases}$$

$$\text{Given } PQ = 5 \Rightarrow |y_1 - y_2| = 5$$

$$\Rightarrow 4k^2 + 4k - 24 = 0 \Rightarrow k = 2 \text{ or } -3$$

$$\text{as } k > 0 \quad k = 2$$

Q12

$$\text{Here } \alpha = \frac{-1+\sqrt{3}i}{2} = \omega$$

$$\text{Now, } (2 + \omega)^4 = a + b\omega$$

$$\Rightarrow (4 + \omega^2 + 4\omega)^2 = a + b\omega$$

$$\Rightarrow (\omega^2 + 4(1 + \omega))^2 = a + b\omega$$

$$\Rightarrow (\omega^2 - 4\omega^2)^2 = a + b\omega$$

$$\Rightarrow (-3\omega^2)^2 = a + b\omega$$

$$\Rightarrow 9\omega^4 = a + b\omega$$

$$\Rightarrow 9\omega = a + b\omega, \quad \{\because \omega^3 = 1\}$$

$$\Rightarrow a = 0, b = 9$$

$$\Rightarrow a + b = 0 + 9 = 9$$

Q13

$$D(z - 2\operatorname{Re}(z))$$

$$C(\bar{z} - 2\operatorname{Re}(\bar{z}))$$



$$A(z)$$

$$B(\bar{z})$$

$$\text{Let } z = x + iy$$

$$\therefore \text{Length of side of square} = 4 \text{ units}$$

$$\Rightarrow |z - \bar{z}| = 4 \Rightarrow |2iy| = 4$$

$$\Rightarrow |y| = 2$$

$$\text{Also } |z - (z - 2\operatorname{Re}(z))| = 4$$

$$\Rightarrow |2\operatorname{Re}(z)| = 4 \Rightarrow |2x| = 4 \Rightarrow |x| = 2$$

$$\text{Now } |z| = \sqrt{x^2 + y^2}$$

$$= \sqrt{4 + 4} = 2\sqrt{2}$$

Q14

$$\therefore -1 + \sqrt{3}i = 2 \cdot e^{\frac{2\pi}{3}i}$$

$$\text{and } 1 - i = \sqrt{2} \cdot e^{\frac{i\pi}{4}}$$

$$\text{So, } \left(\frac{-1 + \sqrt{3}i}{1 - i} \right)^{30} = \left(\sqrt{2} e^{\left(\frac{2\pi}{3} + \frac{\pi}{4} \right)i} \right)^{30}$$

$$= 2^{15} \cdot e^{-\frac{\pi}{2}i} = -2^{15} \cdot i$$

Q15

$$\therefore |z| - \operatorname{Re}(z) \leq 1$$

$$\sqrt{x^2 + y^2} - x \leq 1$$

$$\sqrt{x^2 + y^2} \leq 1 + x$$

$$x^2 + y^2 \leq 1 + x^2 + 2x$$

$$y^2 \leq 1 + 2x$$

$$y^2 \leq 2 \left(x + \frac{1}{2} \right)$$

Q16

$$z = x + iy$$

$$x^2 - y^2 + 2ixy = i(x^2 + y^2)$$

$$\Rightarrow x^2 - y^2 = 0 \text{ and } 2xy = x^2 + y^2$$

$$\Rightarrow (x - y)(x + y) = 0 \text{ and } (x - y)^2 = 0$$

$$\Rightarrow x = y$$